5.1. RADIOACTIVE DECAY

Radioactive decay is a spontaneous nuclear transformation that has been shown to be unaffected by pressure, temperature, chemical form, etc (except a few very special cases). This insensitivity to extranuclear conditions allows us to characterize radioactive nuclei by their decay period and their mode and energy of decay without regard to their physical or chemical condition.

The time dependence of radioactive decay is expressed in terms of the *half-life* ($t_{½}$), which is the time required for one-half of the radioactive atoms in a sample to undergo decay. In practice this is the time for the measured radioactive intensity (or simply, *radioactivity* of a sample) to decrease to one-half of its previous value (see Fig. 1.1). Half-lives vary from millions of years to fractions of seconds. While half-lives between a minute and a year are easily determined with fairly simple laboratory techniques, the determination of much shorter half-lives requires elaborate techniques with advanced instrumentation. The shortest half-life measurable today is about 10^{-18} s. Consequently, radioactive decay which occurs with a time period less than 10^{-18} s is considered to be instantaneous. At the other extreme, if the half-life of the radioactive decay exceeds 10^{15} y, the decay usually cannot be observed above the normal signal background present in the detectors. Therefore, nuclides which may have half-lives greater than 10^{15} y are normally considered to be stable to radioactive decay. However, a few unstable nuclides with extremely long half-lives, $\geq 10^{20}$ y, e.g. ^{82}Se, cf §5.4.4, have been identified. It should be realized that 10^{20} y is about 10 billion times larger than the estimated age of the universe.

Radioactive decay involves a transition from a definite quantum state of the original nuclide to a definite quantum state of the product nuclide. The energy difference between the two quantum levels involved in the transition corresponds to the *decay energy*. This decay energy appears in the form of electromagnetic radiation and as the kinetic energy of the products.

The mode of radioactive decay is dependent upon the particular nuclide involved. We have seen in Ch. 1 that radioactive decay can be characterized by α-, β-, and γ-radiation. *Alpha-decay* is the emission of helium nuclei. *Beta-decay* is the creation and emission of either electrons or positrons, or the process of electron capture. *Gamma-decay* is the emission of electromagnetic radiation where the transition occurs between energy levels of the same nucleus. An additional mode of radioactive decay is that of *internal conversion* in which a nucleus loses its energy by interaction of the nuclear field with that of the orbital electrons, causing expulsion of an electron and ionization of the atom instead of γ-ray emission. A mode of radioactive decay which is observed only in the heaviest nuclei is that of *spontaneous fission* in which the nucleus dissociates spontaneously into two roughly equal parts. This fission is accompanied by the emission of electromagnetic radiation and of neutrons. In the last decades also some unusual decay modes

have been observed for nuclides very far from the stability line, namely *neutron emission* and *proton emission*. A few very rare decay modes like ^{12}C-emission have also been observed.

In the following, for convenience, we sometimes use an abbreviated form for decay reactions, as illustrated for the ^{238}U decay chain described in §1.3:

$$^{238}\mathrm{U}(\alpha)\,^{234}\mathrm{Th}(\beta^-)\,^{234}\mathrm{Pa}(\beta^-)\,^{234}\mathrm{U}(\alpha),\ \text{etc.}, \tag{5.1}$$

or, if half-lives and details are of importance:

$$^{238}\mathrm{U}(\alpha,\ 4.5\times10^{9}\ \mathrm{y})\,^{234}\mathrm{Th}(\beta^-,\ 24\ \mathrm{d})\,^{234\mathrm{m}}\mathrm{Pa}(\beta^-,\ \mathrm{IT},\ 1.17\ \mathrm{min})$$
$$^{234}\mathrm{Pa}(\beta^-,\ 6.7\ \mathrm{h})\,^{234}\mathrm{U}(\alpha,\ 2.5\times10^{5}\ \mathrm{y}),\ \text{etc.} \tag{5.2}$$

In the following chapter we discuss the energetics of the decay processes based on nuclear binding energy considerations and simple mechanics, then we consider the kinetics of the processes. In Ch. 6, where the internal properties of the nuclei are studied, the explanations of many of the phenomena discussed in this chapter are presented in terms of simple quantum mechanical rules.

5.2. CONSERVATION LAWS

In radioactive decay – as well as in other nuclear reactions – a number of *conservation laws* must be fulfilled. Such laws place stringent limitations on the events which may occur.

Consider the reaction

$$\mathrm{X}_1 + \mathrm{X}_2 \ \rightarrow\ \mathrm{X}_3 + \mathrm{X}_4 \tag{5.3}$$

where X represents any nuclear or elementary particle. In induced nuclear reactions X_1 may be the bombarding particle (e.g. a ^{4}He atom in a beam of α-particles) and X_2 the target atom (e.g. ^{14}N atoms), and X_3 and X_4 the products formed (e.g. ^{1}H and ^{17}O).

Sometimes only one product is formed, sometimes more than two. In radioactive decay several products are formed; reaction (5.3) is then better written $\mathrm{X}_1 \rightarrow \mathrm{X}_2 + \mathrm{X}_3$. For generality, however, we discuss the conservation laws for the case (5.3).

For the general reaction (5.3):

(a) The *total energy* of the system must be constant, i.e.

$$E_1 + E_2 = E_3 + E_4 \tag{5.4}$$

where E includes all energy forms: mass energy (§4.3), kinetic energy, electrostatic energy, etc.

(b) The *linear momentum*

$$p = mv \tag{5.5}$$